NSD Compendium

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November 2026

1 Introduction

1.1 Overview of the NSD:

The Natural Scenes Dataset (NSD) is a large-scale 7T fMRI dataset designed to investigate visual processing of natural scenes. It stands out for both its unprecedented scale and the comprehensive preprocessing pipeline provided with it. The NSD includes data from eight participants, each scanned across 30–40 sessions over a year, resulting in about 22–24 hours of high-resolution recordings per subject. Participants underwent approximately 135,000 trials, about 22 to 30 thousand per subject, corresponding to 10,000 distinct images drawn from the COCO (Common Objects in Context) dataset per subject, for a total of 70,000 images, 1000 of which were common to all subjects while the others were mutually exclusive.

Stimuli were shown at a relatively large visual angle (8.4°), engaging wide portions of the visual field. Data were acquired at 1.8 mm isotropic resolution with a temporal resolution of 1.6 s, ensuring fine-grained measurements across both early visual areas and higher-order cortical regions.

Beyond the main task of natural visual stimuli processing, subjects also participated in various other tasks involving recognition, temporal memory and resting state. They had to perform a continuous recognition task which involved signaling if the image they were currently viewing was previously encountered at any point during the experiment. They also provided resting state recordings (6 to 35 5-minutes runs each) and performed a temporal memory task in which they were required to place a portion of the images they saw during the recording experiment on a timescale, indicating when approximately that image was first encountered.

Aside from raw imaging, the NSD provides extensive preprocessing, including motion correction, distortion correction, surface-based alignment, and detailed population receptive field (pRF) estimates for each voxel, which enable precise mapping of visual field coverage. Voxel-level beta estimates, anatomical reconstructions, and ROI definitions spanning multiple cortical systems are also provided. NSD−specific tools such as "*nsd_access*" and associated GitHub resources are available and allow for efficient querying and visualization of the data, tightly linking fMRI responses to the rich COCO image annotations (for each image, COCO contains object labels, categories and 5 sentence-descriptors).

Its combination of scale, longitudinal acquisitions, resolution, tools availability, range of tasks and high-quality preprocessing makes the NSD stand out as a unique and unprecedented resource with a wide range of applications and an incredible versatility, making it suitable for both traditional

voxel-wise analyses as well as more modern AI-driven approaches, such as encoding and decoding models and generative reconstructions. It can also be employed in a wide range of projects independently of the approach employed, as it is suitable for both experimental approaches and engineering approaches, as we will see in the next sections.

1.2 Aim and purpose:

This report's aim is to briefly present a selection of studies that have used the NSD as their primary dataset. Purpose is giving an overview of topics, findings, methods and techniques used within the context of the studies being considered. The following studies make use of the NSD as their primary (or one of their primary) datasets. Studies employing it only as a control dataset and studies that were not relevant to the field of neuroscience have been excluded from this report. If multiple studies addressed the same topic and came to similar findings, only one of them was selected and inserted in this review, usually the first one to be published.

2 NeuroGen: Activation Optimized Image Synthesis for Discovery Neuroscience

2.1 Reference

https://doi.org/10.1016/j.neuroimage.2021.118812

2.2 Summary

NeuroGen introduces a novel framework that fuses fMRI-based neural encoding models with deep generative networks, enabling the system to synthesize images predicted to elicit specific brain activation patterns. Training involves mapping synthetic images through an encoding model to predict fMRI responses and optimizing these images to target distinct activation profiles.

2.3 Methodology

Encoding models: Built several encoding models from pairs of natural images and respective fMRI responses (specifically a Deepnet feature-weighted receptive field encoding model).

Frameworks and tools used to build the encoding models: Deepnet-fwRF framework, AlexNet (used as a feature extractor), 2D Gaussian pooling field (applied to the image to obtain a set of multi-scale feature maps), ridge regression uses the multi-scale feature maps to predict activation for individual regions.

Conditional Deep Generative Model: Integrated them with a conditional deep generative network (BigGAN-deep) to create synthetic images starting from the models' predictions.

2.4 Results

Obtained images are optimal to evoke activation in predefined brain regions, and individual differences are taken into account.

Synthetic images heighten region-specific responses.

Synthetic images can maximize or minimize region specific responses, they also have more extreme predicted co-activation of region-pairs or region-groups than natural images, meaning that synthetic images can tailor the activation of more than one region at the same time as desired.

Encoding models accurately map images to brain responses and remove noise present in fMRI activation maps.

The model is able to detect and amplify features that are important in evoking individual-specific and region-specific responses, and these features are reflected in the synthetic images generated.

3 BrainChat: Interactive Semantic Information Decoding from fMRI Using Vision-Language Pretrained Models

3.1 Reference

https://doi.org/10.1109/ICASSP49660.2025.10889434

3.2 Summary

BrainChat is a generative AI system designed to decode semantic content directly from fMRI scans, achieving both caption generation and question-answering. It leverages CoCa (Contrastive Captioner) vision-language models, combined with a novel Masked Brain Modeling (MBM) self-supervision method.

3.3 Methodology

MBM pre-training: Mask parts of sparse fMRI data and reconstruct them, improving latent representation quality

Contrastive alignment: Align fMRI embeddings with image and text embeddings via CoCa.

Generative decoding: Use cross-attention transformer decoders to predict textual content (captions or answers).

3.4 Results

Achieves state-of-the-art performance in fMRI-to-text captioning.

Successfully implements fMRI-based question-answering—first in the field.

Maintains high performance even without visual input, demonstrating real-world robustness.

4 A highly selective response to food in human visual cortex revealed by hypothesis-free voxel decomposition

4.1 Reference

https://doi.org/10.1016/j.cub.2022.08.009

4.2 Summary

Researchers discovered a component in the visual cortex whose activation is specifically selective to food. They applied a data driven voxel decomposition approach (hypothesis free) and when this selectivity emerged, they ran a series of experiments to rule out that other factors (such as color, shape etc.) were triggering the activation, and found out that activation was correlated with food salience in a picture. The study was conducted in 2 phases, with phase 1 taking subjects 1, 2, 5 and 7 into account, while the remaining 4 subjects were addressed in phase 2.

4.3 Methodology

The matrix of response magnitude of each voxel to each stimulus was factorized into a set of components, which most likely correspond to distinct neural populations, using Bayesian non-negative matrix factorization.

Each component is characterized by a response profile across stimuli and a weight matrix indicating how strongly that component contributes to each voxel's response. (basically after matrix factorization we obtained 2 matrices, a response profile matrix, indicating for each stimulus the activation of each component, and a weight matrix, indicating for each component its contribution in each voxel activation).

The optimum number of components was chosen as the minimum BIC (bayesian information criterion).

Shared components were isolated by measuring pairwise inter-subject correlation of component response profiles across shared stimuli (515 images in phase 1, 1000 in phase 2). 5 components were consistent across subjects.

For each component, they inspected the top 25 stimuli (25 images yielding highest response magnitude for that component). For component 3 they were all portraying food.

To ensure that component 3 was actually selective to food and rule out that other factors were triggering the activation, they ran other experiments, mostly calculating correlation between component activation and the presence of other image features and comparing those with correlation with food salience.

Built a CNN-based encoding model of Component 3 response to enable tests on images beyond those in the NSD. Top 50 images were food.

Repeated analysis on BOLD5000.v2 dataset, and found similar results.

4.4 Results

Discovered a distinct neural population in the ventral visual pathway whose activation is specifically selective to food. They named it ventral food component.

5 Encoding of Visual Objects in the Human Medial Temporal Lobe

5.1 Reference

https://doi.org/10.1523/JNEUROSCI.2135-23.2024

5.2 Summary

This study focuses on figuring out how the MTL (medial temporal lobe) encodes visual objects through three different neural coding models—semantic (category-selective), axis-based coding (visual feature-selective), and region-based coding (receptive field-selective). They examined how these coding schemes are distributed across the MTL and how they interact with each other, with the aim of delineating how the information flows within the MTL.

5.3 Methodology

Feature extraction: employed a DNN (specifically an implementation based on the ResNet convolutional Neural Network architecture) to extract features for each image.

Feature space construction: employed t-SNE to transform high dimensional features into a two-dimensional feature space. This was used to identify region-based coding. Tested robustness by varying the perplexity parameter of t-SNE and examining the resulting number of region-coding voxels. This number was stable so it proved results were robust to the perplexity. Then applied PCA to reduce high-dimensional features, this also allowed feature distribution visualization. resulting feature space was used to identify axis-coding voxels.

Voxel Selection: The methods employed to identify relevant voxels differed based on the type of voxel, as follows:

1. Semantic-coding voxels were identified using a 1×20 ANOVA to detect significantly unequal responses across object categories, with an additional threshold requiring responses to be at least 1.5 standard deviations above the mean.

2. Axis-coding voxels were detected using Partial Least Squares (PLS) regression on the DNN features, selecting four components via 10-fold cross-validation to explain at least 80% of variance.

3. For Region-coding voxels, a response density map was estimated within the t-SNE feature space using a 2D Gaussian kernel, and permutation tests ($p < 0.01$) were used to identify significant receptive regions; voxels responding preferentially to images within such regions were classified as region-coding voxels.

Statistical Analyses and Metrics: The Degree of Selectivity (DOS) Index and Response Ratio quantified category selectivity. Representational Similarity Analysis (RSA), using Spearman correlation, compared the representational structures between subregions. Pairwise distance metrics evaluated the relationship between coding models (axis vs. region coding). Finally, population decoding with a Maximum Correlation Coefficient (MCC) classifier, using z-scored voxel responses, was applied to predict object categories.

5.4 Results

semantic coding was found across all subregions of the MTL, axis coding only in the PHC (parahippocampal cortex) and PRC (perirhinal cortex), region coding primarily in the PHC and PRC.

proportion of different coding voxels changed across MTL subregions. Semantic coding voxels increased from PHC to amygdala, while the proportion of axis coding voxels decreased.

Ratio of semantic coding voxels over axis coding voxels increased monotonically from PHC to amygdala.

6 Retrieving and reconstructing conceptually similar images from fMRI with latent diffusion models and a neuro-inspired brain decoding model

6.1 Reference

https://doi.org/10.1088/1741-2552/ad593c

6.2 Summary

This study proposes a novel approach to brain decoding that relies on semantic and contextual similarity, rather than on image features as most of the other approaches do. They created a deep learning pipeline that includes a linear brain-to-feature model that maps fMRI activity to semantic visual stimuli features, they assume the brain projects visual info into a space homeomorphic to the latent space of last layer of a pretrained neural network, these features are categorized in the latent space and results are used to retrieve images or conditions to pass to a generative latent diffusion model to generate new images. They employed NSD, BOLD5000 and GOD datasets.

6.3 Methodology

Re-labeling of NSD and BOLD5000: they relabeled each image in order to introduce semantic pseudo-labels. The process was the following:

1. computation of CLIP CLS embeddings for each image

2. K-means clustering, based on CLIP embedding similarity, 25 was the optimal number of clusters.

3. resulting clusters effectively served as semantic pseudo-labels.

fMRI+KNN to perform image retrieval on the datasets, basically aim was predicting the cluster an image belongs to from fMRI data, then use KNN to retrieve K most similar images.

Subject-specific encoding models: One for each subject, they were built assuming that the brain processes inputs employing a hierarchical approach similar to CNNs or CLIP models. They propose a linear mapping between fMRI data and the last convolutional layer of a pre-trained frozen CLIP architecture. The model takes as input an activity pattern and outputs a feature vector h which is an estimate of the image features, therefore is directly comparable with CLIP embeddings.

Stimulus classification: they run the KNN algorithm in the latent space with k=5 and assign it to the class corresponding to the most frequently occurring label out of the 25.

Image Retrieval: To retrieve an image, they simply run KNN again, with k=3, in order to identify the closest matches between the estimated image features (h) and the actual image features (CLIP embeddings). The distance metric used is Euclidean Distance.

Generative decoding: To reconstruct the stimulus, they use the predicted label as conditioning for reconstructing images. In this study, they used a pretrained image generator belonging to the family of denoising probabilistic diffusion models. For GOD dataset only.

6.4 Results

Successfully retrieved same or same-category images in largely most cases. (Accuracies: GOD Top-1: 20%, GOD Top-3: 40%, BOLD5000 Top-1: 57%, BOLD5000 Top-3: 82%, NSD Top-1: 72%, NSD Top-3: 88%)

For the GOD dataset, they successfully retrieved the same or a similar image even for the visual imagery part, and while results were significantly lower (since imagination and vision only share some common processes with vision) they are still above chance level (Top-1: 8%, Top-3: 20%, chance levels were 2% and 6%)

When it comes to image generation, the approach presented yielded a FID score of approximately 10.6. Generated images are not a pixel-perfect reconstruction of the original stimulus, but are semantically close to the original (a swan has been reconstructed into a parrot for example, but still both are birds).

7 Unraveling the Differential efficiency of Dorsal and Ventral Pathways in Visual Semantic Decoding

7.1 Reference

https://doi.org/10.1142/S0129065725500091

7.2 Summary

This study's aim is to investigate the distinctions between dorsal and ventral pathways in semantic decoding. They built a GNN-based decoding model (consisting of a Visual encoder based on

GRU, a label encoder based on GloVE and a GNN-based decoder) through which they investigated decoding differences between the two pathways. Their results showed that the dorsal pathways had significantly higher decoding accuracies for verbs and for nouns with motion attributes than the ventral pathway, while no significant difference was found for other categories. Significant differences in decoding between the two pathways were observed in higher-level visual areas, not in low-level ones. Lastly, decoding accuracies for verbs and for nouns were similar, in spite of the fact that the sample size for nouns was much larger than that of verbs.

7.3 Methodology

ROIs delineation: Regions of Interest were delineated using the "cortical surface probabilistic atlas" template, proposed by Wang et al. In total, they selected 17 regions as ROIs, 8 for the ventral pathways and 9 for the dorsal pathway.

fMRI signal extraction: Based on ROIs, the extraction process followed the following steps:

1) ventral and dorsal pathway regions were denoted as "V" and "D",

2) "V" and "D" were further divided in low-level regions and high-level regions, denoted as "LV", "HV", "LD" and "HD".

Now they had a total of 6 combined regions, and decoding models were trained accordingly.

Semantic label extraction from descriptions provided in COCO dataset: stimuli of the NSD dataset come from COCO dataset, which includes 330k images, for each image there are categories, labels (only nouns, not ok for this study) and 5 sentence-descriptors. Labels were extracted as follows:

1) For each image, words occurring two or more times in the descriptors were extracted (provisional labels).

2) For all images, the occurrences of each word were counted and words below a frequency threshold were removed.

3) Each word was classified into one of three classes: nouns, adjectives or verbs.

4) Nouns were then split into 5 subcategories (people, sports, traffic, indoor, outdoor) and adjectives into 3 (shapes, colors, position).

5) To the nouns, a motion/motionless attribute was added, objects were classified on whether they're commonly found in a motion state in daily life, rather than on whether they possess active motion attributes.

Model: They utilized a semantic decoding model to predict the semantic labels of stimulus images from the neural activity recorded in the six combined cortical regions. The decoding model comprises two encoders and a decoder. The encoders are a GRU-based visual encoder, which takes as input images and activity and outputs visual features, and a label encoder based on GloVE, which takes

as input the descriptors, extracts the labels and outputs word embeddings. The decoder is a GNN-based decoder, which takes as input the outputs of the 2 encoders, makes use of a co-occurrence matrix and outputs a prediction probability for each semantic label.

Statistical Analysis Methods and experiments with semantic decoding models: for each participant (8 subjects of NSD), average decoding accuracy was calculated. Among subjects, the decoding accuracies for both pathways were calculated for each semantic category. To assess the significance of decoding differences, they used the Paired Comparison Wilcoxon nonparametric test. In order to verify the robustness of their results, they employed 2 other representative deep learning frameworks as comparative decoding models: a GRU model and a simpler DNN model.

7.4 Results

Semantic information was successfully decoded in both pathways, for all six combined regions, average accuracy was above chance level (0.015), with "D" and "V" exhibiting highest accuracies, followed by High-level areas "HV" and "HD", followed by low-level areas "LV" and "LD" exhibiting the lowest accuracies.

Both pathways process object attributes similarly, but the dorsal pathway outperforms the ventral in processing motion information. Plus, the dorsal pathway has a more comprehensive ability in visual info processing, likely due to its role in simultaneously determining object attributes and processing motion.

Differences between ventral and dorsal pathway were more pronounced in the high level visual cortex than in the low level one, as "HD" outperformed "HV" in decoding accuracy of verbs, nouns with motion, People and Sports, while between "LV" and "LD" there was no significant difference in decoding performances.

Decoding accuracy of nouns was no better than that of Verbs, in spite of the fact that the former appeared more frequently than the latter in the training samples. Actual accuracy rank was: Nouns>Verbs>Adjectives, while, based on frequency distribution of semantic labels, the expected rank was: Nouns>Adjectives>Verbs, also, decoding accuracy of Nouns was better than that of Verbs only for HV.

All findings were observed in all 3 models, therefore the comparison suggests that results are stable regardless of model framework.

8 Situating the salience and parietal memory networks in the context of multiple parallel distributed networks using precision functional mapping

8.1 Reference

https://doi.org/10.1016/j.celrep.2024.115207

8.2 Summary

Recognition is often divided into recollection and familiarity. Familiarity tasks activate the PMN (Parietal Memory Network), while recollection tasks are often carried out by a broad network that includes regions in the canonical DN (default network). The DN is a widely distributed network containing regions in multiple association areas, and this organizational motif is characteristic of association cortex. In contrast, the PMN and the SAL (salience) networks, although also located deep within association zones, do not follow this motif. The SAL network is restricted to anterior cortices, the PMN to posterior ones and both of them should be more distributed. Therefore, the authors of this study hypothesized that the PMN and the SAL are actually the anterior and posterior components of a larger unified system that fits the distributed network motif. Using high-resolution imaging, they showed that the PMN actually encompasses frontal and insula regions canonically ascribed to SAL, therefore proving the existence of a unified widely distributed network, which they rename SAL/PMN, instead of two distinct restricted ones.

8.3 Methodology

Seed-based functional connectivity: Seeds were placed in both canonical PMN regions (PCU, rPCC) and in frontal regions (rPFC, mPFC). Correlation maps consistently revealed a distributed network beyond the posteromedial cortex.

Clustering approaches: Multi-session hierarchical Bayesian model (MS-HBM) and k-means clustering were applied to define networks in a data-driven manner. Results were compared with seed-based analyses to confirm robustness.

Cross-dataset replication: Analyses were repeated in an independent 3T multi-echo dataset and in the UK Biobank (4,181 participants). Subcortical seeds (ventral striatum, posterior MTL) were also tested to probe connectivity patterns.

Statistical tests: Within-network vs. between-network correlations were compared (paired t-tests, corrected). Surface area analysis (ANOVA, Tukey post-hoc) assessed whether SAL/PMN matched the expected size of association networks.

8.4 Results

The PMN extended across at least 9 cortical zones (PCU, rPCC, RMC, mPFC, vmPFC, rPFC, dPFC, aINS, IPL), overlapping with canonical SAL regions.

Strong functional connectivity linked anterior (SAL) and posterior (PMN) regions, supporting the existence of a single distributed network.

SAL/PMN was found to be closely interdigitated with, but statistically distinct from, other nearby large-scale networks, specifically, the default network, frontoparietal control, and cingulo-opercular networks. Notably, it showed closer correspondence with frontoparietal control networks (both A and B) than with canonical DN regions.

The entire SAL/PMN shows evidence of task engagement in recognition tasks typically associated with PMN only.

Findings were robust across 7T NSD data, 3T independent dataset, and UK Biobank analyses. Higher-resolution imaging preserved the unified SAL/PMN more consistently than 3T, supporting the hypothesis that it had been previously missed due to poor imaging resolution.

Both anterior and posterior regions of SAL/PMN exhibited a significant repetition enhancement effect during recognition tasks. Other networks (DN-A, DN-B) instead showed repetition suppression.

The unified SAL/PMN matched the 4–8% cortical surface area typical of association networks, whereas when treated separately, the SAL and the PMN accounted for about 2% of the surface area each.

9 Brain-optimized Deep Neural Network models of Human visual areas learn non-hierarchical representations

9.1 Reference

https://doi.org/10.1038/s41467-023-38674-4

9.2 Summary

Many DNNs that have been trained to solve computer vision tasks can accurately predict the human brain's responses to real-world visual stimuli. One hypothesis for this success is that it depends in part on the fact that DNNs end up implementing a hierarchy of visual processing stages. To test this hypothesis, the authors of this study implemented 3 DNNs, an AlexNet pre-trained to classify objects in the ImageNet database (computer vision task), and two DNNs directly optimized to predict human brain activity in the main 4 visual areas (V1, V2, V3 and V4, both ventral and dorsal pathways), a single-branch one was trained to predict activity in all four areas jointly, and a multi-branch one was trained to predict activity in the four areas independently. They found out that, while only two of the three models learnt hierarchical representations, the predictions accuracy yielded were similar. Hierarchical representations are therefore not necessary to predict activity.

9.3 Methodology

ROI extraction: ROIs have been drawn based on the results of a pRF (population receptive field) retinotopic mapping experiment. Across all 8 subjects they extracted in total 29597 voxels across all subjects and for all 8 ROIs (V1v, V1d, V2v, V2d, V3v, V3d, V4v, V4d).

General Encoding Model Architecture: Encoding models (both for the task-optimized and the brain-optimized networks) consisted of a feature extractor (a DNN) and multiple read-out heads. Feature extractors are always sequences of transformations operating on an input image x. Each layer applies a transformation on the output of the previous layer. Read-out heads convert the output of the feature extractor into prediction of the brain activity for each voxel.

Task-optimized model: The feature extractor for this model was an AlexNet Deep convolutional Neural Network. 512 feature maps were selected for each layer that exposed more than 512 of them, the final model therefore exposed 2688 feature maps. They constructed two variants of the AlexNet

model, one with a Gaussian pooling field (AlexNet-gpf) and one with flexible pooling field (AlexNet-fpf), for this second one, the read-out heads included multiple pooling fields. For both variants, feature maps from each layer and throughout the depth of the feature extractor were input to the read-out head. For both models the feature extractor was pre-trained, therefore only the parameters of the read-out heads were optimized. For the AlexNet-gpf, the three pooling field parameters were learnt via grid search over a list of 2680 candidates, while for the AlexNet-fpf, read-out heads were trained using gradient descent with the ADAM optimizer. This model was used as a baseline.

Brain-optimized model: the feature extractor for these networks is referred to as a GNet. It consists of a pre-filtering network (taken from a task-optimized AlexNet) followed by a Deep feature extractor. The pre-filtering network takes as input a 227x227 image and outputs an embedding of 192 features at a 27x27 spatial resolution. The deep feature extractor is a sequence of blocks of layers, each block consisting of a batch normalization layer, a dropout layer and a convolutional layer that preserves feature map resolution. For the GNet only flexible pooling fields were used: feature maps of the same resolution were concatenated and a single spatial pooling field was learnt for each spatial resolution, thus, the read-out heads contained multiple pooling fields. A fully differentiable nonlinearity was applied either before or after spatial pooling, or both. Both the feature extractor and the read-out heads have been trained jointly using gradient descent with the ADAM optimizer. Two GNet models were trained, a single branch (GNet8j) and a multi-branch (GNet8r).

Gabor encoding model: The Gabor model feature extractor consists of a single set of fixed convolutions. The read-out head had a single Gaussian pooling field, and the model was trained using grid search over the pooling field parameters with the same candidate grid used for the AlexNet-gpf model, followed by ridge regression to determine feature weights.

9.4 Results

GNet8j and GNet8r performed similarly, managed to outperform the AlexNet baseline and achieved similar prediction accuracies.

AlexNet and single-branch GNet learnt all three types of hierarchy considered in this study (compositional, entailment and integration hierarchies), while the multi-branch GNet learnt only integration hierarchy.

Results indicate that compositional and entailment hierarchies are not necessary for accurate prediction of brain activity, since only the two single branch architectures expressed these kinds of hierarchies, this type of architecture enforces a correlation between integration and compositional hierarchies.

Integration hierarchy might be a special case of a "visual representation gradient" (a property of visual representations that changes monotonically with progression along a visual stream).

This work, while focusing mainly on hierarchies related to representations, suggest that also the relationship between DNN architecture and hierarchies at the level of cortical anatomy might be tenuous, as single-branch architectures are compatible with some kinds of anatomical hierarchies, while multi-branch is not, and yet, both these architectures yielded similar results.

10 Natural Scene reconstruction from fMRI signals using generative latent diffusion

10.1 Reference

https://doi.org/10.1038/s41598-023-42891-8

10.2 Summary

In this study the authors investigated the application of latent diffusion models for brain decoding. They created a two-stage scene reconstruction model called "Brain-diffuser", which in the first stage reconstruct low level properties of the stimulus image from fMRI signals utilizing a VDVAE model, while in the second stage they use an image-to-image framework of a latent diffusion model conditioned on the features predicted during the first step to generate a reconstruction of the initial input. This approach outperformed previous approaches both qualitatively and quantitatively, and when applied to synthetic fMRI patterns generated from individual ROIs it can generate high quality ROI-optimal stimuli.

10.3 Methodology

Stage 1: Since VAE is not sufficient for Natural scenes, they used a VDVAE which is a hierarchical VAE and was introduced by Child, R. in 2021, which trained on a 64x64 resolution ImageNet dataset and had 75 layers of which they used only 31. The training proceeded as follows: they fed images into the encoder and extracted latent variables for each image, then concatenated these latent variables from all the 31 layers and obtained a vector which was then used to train a ridge regression model between fMRI training patterns and the vector itself. The testing consisted of providing the test fMRI patterns to the trained model and then predict latent values for each image, then these latent values were fed to the decoder and images were reconstructed. These were low-level reconstructions which served as an initial guess for the diffusion model.

Stage 2: They trained 2 regression models, one between fMRI and CLIP-Vision features and one between fMRI and CLIP-Text features. Afterwards they encoded the image reconstructed from stage 1 using a AutoKL encoder, added noise and fed this noisy latent of the initial guess to a versatile LDM (a LDM with multiple pathways that allow for conditioning the generation process) conditioned on CLIP-text and CLIP-Vision features, which denoised it. It was then passed through an AutoKL decoder to produce the final reconstruction.

10.4 Results

Their approach was successful in capturing semantics and layout of the ground-truth images.

Brain-diffuser outperformed previous models in both high-level and low-level metrics

It was able to reveal ROI-optimal stimuli by analyzing the reconstructions of synthetic fMRI patterns created from corresponding ROI masks (they simply put the signal from the ROIs they were not interested in to 0, then since these signals were synthetic and therefore out-of-distribution, they re-normalized the latent variables generated from these patterns).

11 Natural scene sampling reveals reliable coarse-scale orientation tuning in human V1

11.1 Reference

https://doi.org/10.1038/s41467-022-34134-7

11.2 Summary

This study addresses a long standing debate in neuroscience by proving that orientation tuning is detectable in BOLD measurements beyond the effects of stimulus vignetting, to which the analysis approaches used to peer past the resolution limitations of standard fMRI measurements have been found to be sensitive. They created two models, a "full" one and a "constrained" one, while the full has orientation-selectivity, the constrained one doesn't. They also revealed a coarse-scale map of orientation selectivity, as preferred orientation varies smoothly across the visual angle.

11.3 Methodology

ROIs: V1, V2, V3 and V4 have been considered as ROIs, although results mainly refer to V1, except for surface maps, which show all regions.

Models: They built 2 models based on the steerable pyramid. The full model simulates each neuron in V1 and has receptive fields that are tuned for both spatial frequency and orientation, and allows for variable weighting of these model neurons. The constrained one also simulates population of neurons in V1, but enforces equal weighting of model neurons across orientation by summing across orientation subbands of the pyramid. The base of both models is a steerable pyramid with 8 orientations, 7 spatial frequency levels and a spatial bandwidth of one octave. The pyramid (and therefore the full model) had a total of 56 filters, after summing across the 8 orientation filters, the constrained model had 7 filters.

Inferring preferred orientation and spatial frequency: they simulated an electrophysiology experiment, which consisted in probing the model with gratings at different orientations and spatial frequencies and measuring its predicted response. The preferred spatial frequency of a voxel was then estimated by averaging the responses across orientations and then computing the mean frequency, weighted by response amplitudes. Similarly, the preferred orientation of a voxel was estimated by averaging across frequencies and then computing the circular mean, weighted by response amplitude.

Cortical surface maps creation: To create a group map, they firstly transformed each subject's data from volume space to surface space using nearest neighbor interpolation, then, all subject's data were transferred into a single cortical space. They then computed the circular mean across subjects for each vertex in V1-V4, weighted by the full model's R square values.

11.4 Results

Constrained and Full models account for roughly the same amount of variance. The full model had a mean R square of 0.0311 and a median R square of 0.0204, while the constrained model had an R square of 0.0310 and a median R square of 0.0212. While these results seem to indicate that there is no reliable orientation-tuning to be modeled, there is also the need to take into account the

fact that the full model has many more parameters than the constrained one, and therefore likely results into overfitting for voxels with low SNR, but produces way higher R squares for voxels with high SNR. Summary statistics do not provide a complete picture.

Model performance depends on the quality of the pRF estimate. The higher the pRF R square, the higher the model's R square. It was found that voxels with high SNR have high-quality pRF estimate and good performance, therefore the model performance depends on SNR, and pRF R square is a good proxy for voxel-wise SNR. The amount of additional variance the full model explained was also correlated with pRF R square.

By plotting voxel orientation preference, they found that there's a clear coarse-scale map of orientations. They found that orientation varies smoothly across the visual angle, as voxels with nearby pRF have similar orientation preference. Also, in V1 it was possible to see a clear progression of orientations that mimics the organization of visual field angle.

Orientation selectivity in V1 is approximately organized in a radial map. The orientation map they created resembled a radial map, so they tested it, also against other two candidates maps: Vertical and cardinal. For each of the three maps, they calculated the deviation between actual voxel's orientation preference and the reference orientation predicted by an ideal version of the map they were testing for. The radial map was the one with lower deviation. They also found that deviation from the ideal radial map was smaller at the meridians and linearly correlated with angular distance with closer meridian. This appears to be a feature of the orientation map

12 Personalized visual encoding model construction with small data

12.1 Reference

https://doi.org/10.1038/s42003-022-04347-z

12.2 Summary

Building personalized encoding models generally requires a large amount of fMRI data for the model to be able to achieve optimal accuracy. The authors of this study propose an ensemble approach to create models for novel individuals with relatively little data, basically by modeling each subject's response vector as a linear combination of the other subjects' predicted response vectors. They showed that their method achieved accuracy comparable to that of models trained on 20,000+ image-response pairs, while still maintaining a patterns of inter-individual differences in the image-response relationship.

12.3 Methodology

ROIs: They considered 4 areas, an early visual area, V1v, and three late visual regions, FFA1, EBA and PPA.

Encoding models: They constructed 5 encoding models for each area for the NSD subjects and 4 encoding models for each area for the NeuroGen subjects (a dataset with 6 participants and 800

image-response pairs for each). For the NSD subjects, they built an individual-20K model, trained on all available data for that subject, a scratch model, trained only on a subset of the total data available, a finetuned model, same as the scratch but with linear readout weights initialized using the average of the individual-20K model weights from the other 7 subjects, a linear ensemble model, which fits a linear model to predict the 8th subject responses from the other 7 subjects' individual-20K model predictions, and lastly an average ensemble model, which predicts the 8th individual responses as the average of the other 7 subjects' individual-20K model predictions. For the NeuroGen dataset, since they didn't have enough data available, they couldn't build the individual-20K model.

Statistical Analysis: To evaluate the model's performance they used prediction accuracy, calculated using Pearson correlation between predicted measured activation, prediction consistency, calculated using Pearson correlation between inter-subject correlation (ISC) of predicted activation (correlation between predictions of each pair of subjects) and ISC of measured activation (correlation between measurements of each pair of subjects).

Feature Extractor: instead of using a fixed, pre-trained FE as most other studies did, they fine-tuned one, with the aim of achieving more accurate encoding models. They used a feature extractor taken from ResNet50, which extracts both low-level and high-level image features using convolutional blocks, a global max pooling layer and a linear readout layer which maps features to brain responses.

12.4 Results

The individual-20K model was of course the best performing. It achieved good accuracy for all the four areas. Top 10 images for each region reflected expected properties associated with activation in these regions, and while there were common top images across subjects, quite a bit of variability was also observed.

For the NSD, both ensemble models outperformed scratch and fine-tuned models. The accuracy of the scratch, fine-tuned and linear ensemble models increase with training data size and with 100+ image-response pairs, the linear ensemble model performs at the same level of the 20K for any region ($p>0.05$, no significant difference in performance).

For the NeuroGen dataset, despite experiencing some (expected) Drop in performance when compared to the NSD, prediction accuracies still remained at a good level. Both ensemble models performed better than the scratch model and for PPA also better than the fine-tuned. Since NeuroGen also contains synthetic images, they tested for differences between the prediction accuracy for synthetic versus natural images, and they found no significant difference.

All models outside of the average ensemble model preserve inter-individual differences, while the best performing in absolute terms is the individual-20K (as expected), the linear ensemble model out of all the others is the one that offers the best balance of accuracy and preservation of inter-individual differences, achieving accuracy similar to the individual-20K while using a training dataset that is only 1.5% of the individual-20K's training data. They also noticed that by adding subjects the accuracy for the NSD improved only for the first two subjects added, no additional subjects was

useful, and for the NeuroGen didn't improve at all, while preservation of inter-individual differences improved significantly with each subject added for both datasets.

13 Human brain responses are modulated when exposed to optimized natural images or synthetically generated images

13.1 Reference

https://doi.org/10.1038/s42003-023-05440-7

13.2 Summary

This study investigates selectivity and inter-individual differences in brain responses to images via fMRI. They conducted two experiments, a group-level one and an individual-level one. They built 8 individual level encoding models for 3 ROIs (FFA1, EBA, VWFA1) and a group average encoding model, which was obtained by simply averaging the 8 individual ones. Based on the Group encoding model, they generated synthetic images and selected top and worst natural images from the NSD for each region. They enrolled 6 new subjects and collected their responses to these images via fMRI, then they applied the ensemble model from their previous study (see paragraph 11) to build personalized individual level encoding models for each of these new subjects. They then obtained sets of individual-specific natural and synthetic images (same generation/selection procedure as before) and collected their regional responses to these personalized images during a second fMRI scan. Results from the first experiment showed that images predicted to achieve maximal activation evoked better response than images predicted to achieve average activation, and that FFA1 and aTLfaces (both of which are areas whose activation is triggered by faces) had higher activation for synthetic images than for natural images. Results from the second experiment confirmed aTLfaces preference for synthetic images and confirmed effectiveness of using individual-level encoding models over group-level or other individual encoding models for synthetic stimuli generation.

13.3 Methodology

Encoding models: They constructed an encoding model for each of the three region selected as ROIs for each of the 8 NSD subjects. They used a deepnet-fwRF architecture, which as an ImageNet-pretrained AlexNet feature extractor, a gaussian pooling field and a Ridge Regression to map features to brain regional responses. From these encoding models, by averaging them, they created a Group encoding model. They used this model to select the sets of stimuli for the first experiment. Personalized encoding models for each of the new 6 subjects, focusing on FFA1 only, were constructed via the linear ensemble approach used in the previous study (see paragraph 11), where each subject's predicted regional activation is the weighted sum of the predictions from 8 NSD FFA1 models plus a bias term. Linear weights and bias were fit using each subject's image-response pair data from session 1.

Stimulus sets: For the first experiment, they generated 80 * 3 synthetic images, 40 predicted to achieve maximal activation in a specific region (GroupMaxSyn) and 40 predicted to achieve minimal absolute predicted activation in a region (GroupAvgSyn), they did so for each of the 3 regions. They selected the top 40 (GroupMaxNat) and worst 40 natural images (GroupAvgNat) for each region from the NSD. both selection and generation were based on the group encoding model.

In total the first experiment had 12 stimulus sets (4 for each of the 3 regions). For the second experiment, they create 6 sets of stimuli for each subject, SelfMaxNat and SelfMaxSyn are sets of natural and synthetic images that maximized activation of FFA1 for that subject, they contain 32 images each, OtherMaxSyn and OtherMaxNat contained best images of another subject and had 20 images each, and lastly GroupMaxSyn and GroupMaxNat contained the Max Group-level FFA1 images from experiment 1, and had 12 images each, for a total of 128 images shown to each subject. Synthetic images were generated during both experiments using the NeuroGen framework (see paragraph 1).

Statistics and metrics: To compare brain regional responses from the generated and natural images, a Linear mixed effect (LME) model was used, and significance was determined using permutation testing. The discrepancy in brain responses from Max vs Avg images was measured using Cohen's d. Encoding model accuracy was calculated as the Pearson correlation coefficient between predicted and measured brain responses from images that were seen at least twice. To compare accuracies between synthetic and natural images they used t-test and Pearson's r.

13.4 Results

First experiment data showed higher activation for Max images and no significant difference between GroupMaxSyn and GroupMaxNat. This pattern was also observed by analyzing the responses in secondary target regions, except in aTLfaces, and FBA1 where activation was higher for GroupAvgNat instead of GroupMaxNat, and for FBA1, mTLbodies and mTLwords, where activation where higher for Avg images than for Max images for both natural and synthetic images. GroupMaxSyn achieved higher activations than GroupMaxNat in aTLfaces and FBA1, while GroupMaxNat achieved higher activations than GroupMaxSyn in VWFA1, VWFA1 and mTLfaces.

They correlated subjects' encoding model accuracy and the Cohen's d to Max and Avg conditions for Natural and Synthetic images and found that more accurate encoding models result in better alignment of observed and targeted brain activation patterns.

Second experiment data showed that the personalized encoding model seemed to work better for the synthetic images compared to the natural images and for the hierarchically later face regions compared to earlier ones. There were higher FFA1 responses for the SelfMaxSyn then for the OtherMaxSyn and higher FFA1 responses for the SelfMaxSyn than for the GroupMaxSyn. mTLfaces and aTLfaces had higher activation for SelfMaxSyn than for OtherMaxSyn and GroupMaxSyn, while FFA2 only for OtherMaxSyn. They also observed a preference for Synthetic images in aTLfaces and a preference for natural images in the mid-level FFA2 region.

14 The Human Social Cognitive Network contains multiple regions within the amygdala

14.1 Reference

https://doi.org/10.1126/sciadv.adp0453

14.2 Summary

The default network is dissociable in 2 adjacent, interdigitated networks, DN-A and DN-B, the authors of this study previously showed that DN-A is involved in tasks of episodic projections, as it is linked with mnemonic and spatial functions of the posterior MTL, which work with the hippocampus to enable mental scenes construction. Given the connection of DN-A to the posterior MTL, they hypothesized that the specialization of the DN-B for social cognitive processes on the other hand, might be due to a connection to anterior MTL regions, which are linked to social cognition, including the amygdala. They find that this is indeed the case, as the DN-B is connected to the amygdala at or near the basolateral complex and medial nucleus.

14.3 Methodology

Functional connectivity mapping: Seed-based correlations from dorsal prefrontal cortex were used to define DN-A and DN-B in both surface and volume space; correlation maps were thresholded and binarized to visualize distinct versus overlapping regions.

Model-based network estimation: After DN-A and DN-B were defined using FC from manually selected seeds, network definition was repeated using a data-driven multisession hierarchical Bayesian model (MS-HBM) In order to produce converging estimates of the network and validate its definition.

Replication and triplication: they excluded 2 participants from the NSD due to excessive head motion, each of the remaining participants provided 6 to 35 5-minutes runs of resting state data, for participants providing 12 or more runs, data were divided into discovery, replication and triplication datasets, in order to ensure test–retest reliability.

Task-based fMRI at 3T (control): False belief and emotional pain tasks (theory of mind) and scene construction tasks (episodic projection) were analyzed to test functional specialization of DN-A versus DN-B.

Anatomical validation of amygdala regions: Automated segmentations of amygdala nuclei were compared with DN-B regions, supplemented by expert hand-drawn medial amygdala (MeA) masks; Dice coefficients quantified overlap.

Seed-based subcortical confirmation: Seeds placed directly in the basolateral and medial amygdala, as well as the entorhinal cortex, were tested for whether they reproduced DN-B connectivity.

Group-level replication: Independent component analysis (ICA) of resting-state data from the UK Biobank (N = 4181) was used to confirm that DN-B included amygdala and subiculum regions at the population level.

14.4 Results

Two interdigitated networks within the canonical default network were identified: DN-A anchored in the posterior medial temporal lobe (parahippocampal cortex, subiculum), DN-B in the anterior

medial temporal lobe (amygdala, entorhinal cortex).

Theory of mind tasks selectively overlapped with DN-B, while episodic projection tasks selectively overlapped with DN-A; group-level analyses confirmed a significant interaction between task and network.

Bilateral regions at or near the accessory basal, basal, and sometimes lateral nuclei were consistently found in DN-B, confirmed by segmentation overlap and seed-based analyses.

DN-B extended dorsomedially into the medial amygdala in all individuals; overlap was confirmed by expert-drawn segmentations and reproduced by MeA-seeded connectivity maps.

DN-B contained bilateral entorhinal regions that were distinct from DN-A and reproducible across datasets.

In two participants, DN-B showed posterior subiculum regions adjacent to DN-A subiculum regions, suggesting fine-scale interdigitation along the hippocampal formation.

ICA from the UK Biobank revealed DN-B regions in the basolateral amygdala, medial amygdala, and subiculum, supporting generalizability to large populations.

15 Multiple traces and altered signal-to-noise ratio in systems consolidation: Evidence from the 7T fMRI Natural Scenes Dataset

15.1 Reference

https://doi.org/10.1073/pnas.2123426119

15.2 Summary

There are multiple theories and mechanisms that have been proposed to explain memory consolidation in the brain. Some of these mechanisms are SCT (standard consolidation theory), MTT (multiple-trace theory), and Trace Transfer, which is derived from SCT. In this study, the authors examined BOLD correlates of image recognition through the scope of multiple consolidation theories. They found that early and late stage image recognition associates with both MTL and visual cortex. Supporting MTT, they found that MTL contribution to recognition persists for over 200d. They also found supporting evidence for synaptic desaturation as the magnitude of BOLD enhancement among surviving memories is positively linked to the rate of removal of competing traces (signal-to-noise ratio increment). Their work favors MTT and desaturation as complementary consolidative memory mechanisms.

15.3 Methodology

ROIs: They selected as ROIs the most commonly considered portions of the MTL: Hippocampus proper (HP), Parahippocampal cortex (PhC), Perirhinal Cortex (PrC) and Entorhinal Cortex (ErC). HP was split into anterior and posterior portions. They also picked 25 Visual System areas and 3 face-selective regions (OFA, FFA1, FFA2).

Memory Model Fits: They built 2 time-dynamical models, one derived according to MTT principle (MTT model) and one derived according to the trace transfer mechanism (memory chain model). The MTT one was based on 3 assumptions: 1) MTL traces expand over time, 2) this expansion rate decays over time, 3) traces are vulnerable to natural degradation or interference (get replaced by new ones). The memory-chain model on the other hand assumes that memory representation in a store declines in strength while trying to induce new representations in higher-level, more permanent stores. Both models were fit to MTL activation in the within-session (same image shown during the same fMRI session) and outside session (same image shown during different fMRI sessions) timescales.

Analysis: To indicate how many times an image had been previously shown to a subject, rep0 (first time They see it), rep1 (second time) or rep3 (third time) labels were used. They applied a linear logistic regression classifier to different groups of features (MTL, VS, MTL+VS). Only rep1 trials were considered. Their metric of interest was mean balanced accuracy, and differences in this metric between feature sets were identified with a mixed-effect model and tested with a two-way, repeated measure ANOVA and post hoc differences were then assessed between feature groupings.

15.4 Results

They compared activations of each MTL region between within-session image recognition and outside-session image recognition among successful rep1 and rep2 trials and found that outside-session image recognition activation was significantly greater in all areas except HP, meaning that MTL activation during recognition increases over time, as MTT suggests.

They applied a multivariate classifier model of BOLD activations to predict successful vs. unsuccessful recognition, this way they could quantify sets of brain regions that optimally contributed to image recognition. They found that the main effect of ROI combination was highly significant: for the outside session recognition condition, the 25 visual ROIs and the 5 MTL ROIs combined showed a balanced accuracy of 59.6%, significantly better than the 25 Visual ROIs alone (58.6%) and then the MTL alone (56.4%). The same was also observed for within session recognition conditions, meaning that MTL and Cortex remain steady across time.

The MTT model, when applied to the outside session timescale performed better than the memory chain model, which performed better when applied to the within session timescale. This result means that the MTT model better explains the prolonged evolution of memory traces.

From the subject-specific derivatives of memory recognition they computed, they found that peak forgetting rate of rep1 images significantly correlates with peak increase in MTL activation of surviving memory traces within PrC and PhC. They also tested the association of outside-session rep1 peak forgetting rate with the increases in PrC and PhC BOLD activation of rep1 trials within a session and found no significant effect.

They performed a 3-way interaction test with a linear mixed-effect model to test whether the change in MTL connectivity differed based on the image features. They focused on face images (and therefore on face areas) and found that the strongest interaction effect happened between 1 to 20 days since the most recent presentation. They also calculated the correlation at this peak of the within session and face versus no face grouping and found that the interaction effect was driven by

a significant decrease in connectivity in non-face image recognition over time.

16 Contrastive learning explains the emergence and function of visual category-selective regions

16.1 Reference

https://doi.org/10.1126/sciadv.adl1776

16.2 Summary

Modular and distributed coding theories have existed for long, but to this day it is still unclear the extent to which they are compatible or opposed. This study presents a reconciling framework, called contrastive coding. They used a self-supervised instance-level contrastive learning objective to train a Deep convolutional Neural Network over a large set of natural images taken from the ImageNet dataset. They show that category-selective regions naturally emerge for faces, bodies, scenes and words, and that lesions in these model units lead to recognition deficits. Furthermore, these pre-identified regions can predict the activity in the corresponding regions of the visual cortex. This study indicates that brain-like functional specialization can emerge without category-specific learning pressures.

16.3 Methodology

Models: Their main model used a self-supervised learning objective, the Barlow Twins algorithm, to train an AlexNet using the ImageNet dataset. The goal of the Barlow Twins is to reduce redundancy in the neural network's learned representations while preserving the informativeness of its features They also built 3 other models, an AlexNet with group normalization layers trained with instance-prototype contrastive learning, a default TorchVision AlexNet trained with supervised loss on ImageNet, and lastly another AlexNet trained on the VGGFace2 dataset, this one trained with cross-entropy loss and SGD for 100 epochs.

Test for emergent category-selective tuning: They designed a procedure that mirrors the one used to find category-selective regions in the brain. They recorded the activation of every model unit in response to the same localizer image set used to identify category selective regions in the NSD, then performed a statistical contrast to identify nonoverlapping sets of model units that were selective for faces, bodies, scenes and words. This procedure was run for every layer of the model. Then they also tested for an alternate standard localized image set and an alternate localizer method which involved identifying units whose mean response to the preferred category is at least twice of the mean response to any other category.

Lesion experiments: Before lesioning, they instantiated a readout mechanism, but instead of letting the model learn a set of fully connected readout weights for each category, they added a sparsity regularization to the readout weights. The regularized linear readout function was trained for 10 epochs. Then they measured the baseline recognition accuracy for each of the 1000 ImageNet categories. They then carried out the main lesioning experiment and lesioned all of the same category-selective units across the model's relu layers, after each type of lesion (all units of the same category were lesioned during the same iteration) they measured the change in recognition

accuracy for each ImageNet category. They then computed the 1000-dimensional cost profiles for each lesion as the difference between unlesioned and lesioned top-5 accuracies for each category and then compared these cost profiles between all pairs of lesions, measured the degree of impairment of the top 10 categories most affected by each lesion types, examined whether the ImageNet categories that most activated a certain category selective unit was also the hardest to recognize after lesioning, performed a control analysis by lesioning randomized groups of units and measuring the impact, and lastly, they tested for a causal link between selective units across the model hierarchy by silencing a specific category selective unit from layers relu1 to relu5 and then measuring the activation levels to the color probe localizer images in layer relu6.

Encoding model: To test the hypothesis that the pre-identified sets of units from a single contrastive model can capture the response profiles of the classical category selective regions in the brain, they built an encoding model by fitting it for each category selective regions using only the subset of the DNN units that had corresponding selectivity, they also required all learned encoding weights to be sparse and positive. They also performed a control analysis using as encoding models the category selective units of an untrained model as well as two models of lower level image statistics (GistPC and Gabor)

16.4 Results

When testing for emergent category-selective tuning, they found small category selective units in each layer, and the overall proportion and relative strengths of selective units tended to significantly increase across the model hierarchy. It was possible to find units with selectivity for each domain even in a single layer, therefore category selective units emerge in the context of a single population code, as operationalized by a hierarchical layer. They found that these results hold when using the alternate localizer image set, when using the alternate localizer method, when using an AlexNet trained with a instance-prototype contrastive learning, when using an AlexNet trained using a supervised 1000-way imagenet classification objective and also when using an untrained AlexNet (randomly initialized AlexNet architecture), although in this last case number and strength of the category selective units were far lower, suggesting that training is necessary for DNNs to develop reliable category-selectivity.

As a result of the main lesion experiment, they observed that lesions to face-selective units led to the strongest deficits for several dog breed categories, as well as object categories containing people (e.g.seat belt), body-selective units lesions led to the strongest deficit for animals and clothing, scene-selective units lesions led to deficits for certain place categories and large objects categories, word-selective units lesions led to deficits in recognizing insect categories. They also observed that the top 10 categories most affected by lesioning of units selective to a certain category went largely unaffected when lesioning of units selective to a different category happened. Also, the more an ImageNet category activated a group of category selective units, the more difficult it was to recognize that category after lesioning those units, meaning that positive activation magnitude is a reliable indicator of functional relevance. Lastly, they found that similarly tuned groups of units have a meaningful functional link across the DNN layers.

Results from testing the encoding model on the NSD dataset demonstrated that the category selective feature tuning that emerges in a contrastive learning model is able to predict brain response structure of all category selective regions. The control analysis they performed showed that the

3 control models were not able to predict brain activity. Overall, results demonstrated that pre-identified unit subsets from a single DNN model can predict the response structure of 11 diverse category-selective regions (FFA1, FFA2, OFA, EBA, FBA1, FBA2, PPA, OPA, VWFA1, VWFA2 were considered as ROIs).

17 Representations in Human primary visual cortex drift over time

17.1 Reference

https://doi.org/10.1038/s41467-023-40144-w

17.2 Summary

Recent studies in mice found that representations of natural stimuli in the visual cortex are not stable. Up to now it was impossible to test if representational drifts occur in the Human visual cortex as well, as no dataset before the NSD met the requisite longitudinal measurements. This study focused on understanding whether this drift also occurs in humans. They built an encoding model to estimate tuning properties within V1 and during each individual session and used it to test whether tuning changed over time. The model goodness of fit dropped with the number of sessions occurring between train and test sessions, supporting the hypothesis of representational drift in the Human visual cortex.

17.3 Methodology

Encoding model: They constructed an image-computable model based on a steerable pyramid with 8 orientation, 7 spatial frequency levels and a spatial frequency bandwidth of 1. To the 56 oriented filters they added 2 filters for the lowest and highest spatial frequency. The output of each filter in the steerable pyramid was sampled by each voxel's pRF by multiplying the 2D pRF with the filter output. The model had 58 sampled outputs per image for each voxel. They modeled the responses of voxels as a linear combination of the matrix consisting of the voxel' sampled outputs for all filters of all images and a constant term times a vector of beta weights term plus noise.

Statistical analysis: They used cross-validated R-square to evaluate the encoding model's goodness-of-fit across sessions. To test whether drift was caused by changes in variance and/or mean, they equalized either the mean or the variance of each voxel's response within each session and then re-computed the cross-session goodness of fit matrix. To examine how images were represented by the entire V1 population, they measured the correlation between representations in different sessions.

Tuning change simulation: They simulated a 1D Gaussian tuning curve with 4 parameters in order to understand how changing a voxel's tuning curve affects its response's mean and STD. For different values of the 4 parameters they computed the mean and standard deviation.

Population response simulation: They simulated population response in order to evaluate whether or not drift accumulates across sessions, as if so population responses should increasingly differ from the original pattern. To do so, they used the model fits from each session and simulated responses to 100 randomly selected images, for each image created a correlation matrix containing

correlations between the simulated population response of each session and those of all other sessions, averaged all 100 single image matrices, obtaining a matrix per subject, and then averaged all entries corresponding to each interval.

Representational Dissimilarity Matrix (RDM): They constructed one to test the hypothesis that responses to individual stimuli change over time, but their representations remain stable relative to other stimuli. Each RDM consisted of distances between all pairs of simulated population responses in the same session, computed as 1-r (Pearson's correlation coefficient), then, between each pair of RDMs, they computed the Spearman correlation.

17.4 Results

The encoding model's goodness of fit dropped with the number of sessions occurring between train and test sessions, indicating a drift, while the generalization across adjacent sessions remained the same across the experiment, therefore the cause of the drift was not a change in the SNR or in measurements reliability, but rather to progressive changes in a neural process.

While testing if the drift is due to changes in the mean and or variance, they found that equalizing the variance had no impact on the drift, while removing the mean response amplitude significantly attenuated it. They also found that response amplitudes exhibited dependencies across sessions. This suggests that drift is primarily due to changes in mean response amplitude across sessions.

During tuning change simulation, they found that only an additive baseline shift would change the mean response amplitude without affecting the variance. Consistent with this observation, they found that baseline coefficients exhibited positive auto-correlations, which indicates a cumulative shift in the baseline across sessions, which the drift reflects.

Cross-correlation between voxels indicated that changes were not uniform across all V1, suggesting that distinct sub regions undergo different changes in mean response amplitude. To test this, they simulated population responses and found in fact a decreasing similarity between population representations as the number of sessions increased, therefore population-level representations drift across sessions.

They hypothesized that while population-level representation to individual stimuli drift, it remains stable relative to other stimuli. To test this they constructed RDM matrices for each session and found that their correlation was stable, supporting their hypothesis.

18 Re-expression of CA1 and entorhinal activity patterns preserves temporal context memory at long timescales

18.1 Reference

https://doi.org/10.1038/s41467-023-40100-8

18.2 Summary

CA1 and ERC areas are known to be of support to temporal memory. How they encode and represent time information and the extent to which they do so is however still unclear. The authors

of this study focused on characterizing the mechanism that preserves temporal memory when events are re-encountered across long time scales. In particular, they focused on testing whether memory precision was associated with the degree to which patterns were re-expressed (and therefore context was reinstated) upon stimulus re-encountering, and found out that this was the case. Furthermore, they noticed that temporal memory signals were not predictors of recognition confidence, which was mainly predicted by distinct MTL expressions.

18.3 Methodology

Statistical Analysis: they firstly correlated the activity patterns of each roi between each pair of exposures to the same image, then, these correlations were Fisher-transform. They obtained a single similarity measure by averaging between these pairwise correlation and then compared these similarity measures for images associated with high versus low temporal memory precision. A permutation test was then used to evaluate the significance of the difference between high and low temporal memory precision. Then, they performed several mixed-effect analyses, one accounting for temporal lag information, one for V1, one using recognition confidence as the dependent variable instead of temporal precision, and lastly, several different ones that included various combinations of exposure pairs, whose performances were then compared with each other.

Image-specificity analysis: They used 2 approaches to assess image-specificity in CA1 and ERC:

1. Intact versus shuffled pattern similarity analysis: they shuffled the E1-E2 mapping within participants such that each E1 image was paired with a different E2 image, then they calculated the pattern similarity and temporal lags of these new pairs. These scores were submitted to a mixed-effect logistic regression model predicting temporal precision. This procedure was performed 1000 times, resulting in a null distribution of beta values for each ROI.

2. Target versus foil pattern similarity analysis: for each image in the temporal memory test (target), they identified control images (foils) according to 2 criterias: 1) the 2 images shared same E1/E2 session number, but not run number and 2) foils had to receive the same memory judgement as targets (correctly classified at all 3 exposures). They computed pattern similarity between target E1 and target E2 (target similarity) and target E1 and foil E2 (foil similarity). Foil similarity was subtracted from target similarity and this difference was submitted to a mixed-effect logistic regression model as a predictor of temporal memory precision.

18.4 Results

Participants were able to place images in their correct temporal context with above chance-level accuracy at group level. Upon inspection of temporal memory accuracy for each participant, it was found that it was above chance level for 7 out of 8 subjects.

CA1 and ERC had significantly greater pattern similarity across exposures for images whose first exposure to the subject was more precisely placed in time by subjects themselves than those whose first exposure was placed less precisely. They repeated the analysis in 3 variants and found that this result remained consistent even when accounting for temporal lag, that this was true for V1 as well, and that pattern similarity in CA1 and ERC were not a predictor of recognition confidence (PHC was), while they were in fact predictors of temporal memory precision.

Upon focusing on finding out which pair of image exposures was most predictive of temporal memory, they found that the pattern similarity in CA1 and ERC between first and second exposure uniquely predicts temporal memory. Effects on prediction were only marginally significant for similarity between second and third exposure and not significant for similarity between first and third exposure, although exposure 1-exposure 3 similarity in the PHC significantly predicted recognition confidence.

They found that temporal memory was predicted by image-specific pattern similarity rather than by generic memory processes or attention states. They controlled for session effect and this finding held, suggesting that temporal context representation in CA1 and ERC can differentiate images within the same session.

19 Low-level tuning biases in higher visual cortex reflect the semantic informativeness of visual features

19.1 Reference

https://doi.org/10.1167/jov.23.4.8

19.2 Summary

Some high-level visual regions are highly selective to semantic categories. However, they exhibit sensitivity to low-level features as well. There are 2 hypotheses that explain this phenomenon. The First theory, partially supported by previous studies, states that category selective regions are biased in favor of low-level image features properties that are informative of their preferred category. Therefore, these biases play a functional role in both learning and categorization. A Second theory, on the other hand, states that semantic category selectivity reflects representations at a high level of abstraction only. This view implies that low-level feature biases do not play a functional role, rather, they reflect a structural constraint and are merely the result of processes such as inheritance of selectivity from earlier visual areas. This study explores both possibilities via two complementary analyses: First, they used the COCO dataset and a Gabor filter to demonstrate association between low level features and high-level semantic categories. Second, they used the NSD and a voxelwise forward encoding model to estimate feature and spatial selectivity of neural population throughout the visual cortex. They found supporting arguments for the first theory, such as systematic biases in category selective regions towards low-level features informative of their preferred category. Biases were also not driven by selectivity for the categories themselves.

19.3 Methodology

COCO datasets: 50,000 images from the COCO dataset were used to analyze the natural scene statistics. Images were annotated with high-level semantic categories (faces, buildings, animate objects, inanimate objects, small objects, large objects, indoor scenes, outdoor scenes).

Gabor Filters: Gabor filters were applied to extract 96 low-level features (12 orientations $\times$ 8 spatial frequencies). This allowed for systematic quantification of orientation and spatial frequency energy across the images and directly linking them to semantic labels.

AlexNet feature extractor: Features were also extracted from the convolutional layers of an AlexNet. These features captured mid-level visual representations and were used to estimate population receptive fields (pRFs) and improve model fitting accuracy in higher visual areas.

Semantic Features: Semantic information was explicitly modeled using both COCO object and "stuff" annotations. Additional labels were derived to capture coarse semantic dimensions such as animacy, real-world size, and scene type (indoor vs. outdoor). Face labels were obtained using RetinaFace detection.

Encoding models: Voxel-wise forward encoding models were fitted to fMRI data from the NSD. Each voxel's response was modeled as a weighted sum of feature activations within its pRF. This framework jointly estimated spatial tuning and feature selectivity for low-level and mid-level representations.

Statistical Analysis: Ridge regression with cross-validation was used to fit voxel-wise models. Partial correlations and decoding analyses assessed the association between features and categories. Permutation tests and FDR correction were applied to establish statistical significance, and residual-based analyses were performed to dissociate low-level feature tuning from category selectivity.

19.4 Results

Different categories were reliably associated with unique Gabor feature profiles. For example, faces and animate objects correlated with diagonal orientations and lower spatial frequencies, while buildings and large inanimate objects were linked to cardinal orientations (vertical/horizontal) and higher spatial frequencies.

Decoding analyses revealed that faces, small objects, and animate objects were more easily decoded from small, central pRFs, while scene-related categories (indoor/outdoor) and large inanimate objects were more decodable from large or peripheral pRFs. Moreover, the lower visual field carried more diagnostic information for several categories, including faces, buildings, inanimate objects, and outdoor scenes.

Category-selective ROIs showed systematic low-level tuning consistent with their functional role. Scene-selective regions (OPA, PPA, RSC) exhibited biases for cardinal orientations and higher spatial frequencies, while face-selective regions (OFA, FFA) were biased toward lower spatial frequencies and curved/diagonal orientations.

Face- and word-selective regions showed foveal biases, while scene-selective regions showed peripheral biases, aligning with the typical viewing demands of these categories. These spatial tuning patterns mirrored the diagnostic spatial statistics of the relevant semantic content.

Even after regressing out semantic category responses, voxels retained significant tuning to orientation and spatial frequency. This indicates that low-level feature selectivity cannot be reduced to category selectivity alone and plays a functional role in visual processing.

In several regions (e.g., OPA, PPA, EBA), the strength of low-level feature tuning depended on

the semantic content of the images, suggesting that low- and high-level selectivity interact dynamically rather than existing as fully separable processes.

Tuning biases were observed not only in well-known category-selective ROIs but also in dorsal and retinotopic areas, indicating that low-level feature biases form a continuous organizational principle across the visual cortex rather than being restricted to specialized modules.

20 A Texture Statistics Encoding Model Reveals Hierarchical Feature Selectivity across Human Visual Cortex

20.1 Reference

https://doi.org/10.1523/JNEUROSCI.1822-22.2023

20.2 Summary

Visual processing is carried out by brain regions organized hierarchically, with the complexity of representation increasing at each level. There are explanations for response properties at the extremes of the hierarchy, but computations happening at the mid-level regions, which are thought to represent features like contour or texture, are not yet well understood. A texture model proposed by Portilla and Simoncelli in the 2000 (P-S model) has been proved to be useful in understanding mid-level features that drive the visual responses, but this model has been largely tested only on synthetic stimuli, therefore it is unknown how well it captures neural responses to natural scenes. This study bridges this gap by testing such models on the NSD. They found that the model achieves good accuracy throughout the hierarchy, with best performance in early and mid-level areas, and that it recovers the hierarchical organization of the visual cortex, with retinotopic areas best described by low-level features and category-selective areas best described by higher-level ones.

20.3 Methodology

Texture Statistics encoding model: They constructed a forward encoding model based on image-computable statistic features. It included parameters that captured the spatial selectivity of voxels and their selectivity for a range of features related to local image structure. Parameters were fit on a voxelwise basis across a large portion of the occipotemporal cortex as well as several functional ROIs such as V1-V4h, OPA, PPA, RSC, FFA and EBA. For each voxel, they computed the R square value. The model included a range of features related to both low and mid level properties of the image. For comparison, they used several other models commonly used for early vision: 2 versions of the GIST model (2x2 and 4x4 spatial grid versions), a Gabor model and the first 2 layers of a CNN, AlexNet. Later, they created a new version of their model, in which the receptive field size was fixed at a single value for all voxels.

Statistical Analysis: permutation test was used to test significance of prediction accuracy. A variance partitioning analysis was conducted to test which features were more critical for the predictive performance of the model. A second variance partitioning analysis in which they considered each of the subset separately was conducted to more precisely determine which texture features were most uniquely predictive of neural activation.

PCA: They performed PCA on the unique variance values for each of the 10 feature subsets in order to further explore the representational space defined by the model.

20.4 Results

The model achieved good performance across multiple visual areas, particularly in the early visual cortex. Performance declines from V1 through higher category selective areas, but still performance remained relatively high through the visual hierarchy.

The model showed performance comparable to that of other widely used models, with accuracy exceeding that of the 2x2 GIST model, of the Gabor model and of the first layer of the Alex Net, while the 4x4 GIST and the 2nd layer of the AlexNet slightly outperformed it.

The first variance partitioning analysis revealed a gradient from lower to higher visual areas, voxels in the most posterior portion of the occipital cortex tended to have more unique variance explained by lower-level features, while more anterior regions had progressively more variance explained by higher-level features. Still, across all visual areas, a substantial portion of the variance was shared between the 2 groups of features, suggesting some degree of overlap between the 2 feature spaces.

Through the use of a version of the model that allowed for fixed receptive field size, which they fixed at a middle range value of 1.48 degrees, they replicated the key findings found through the original model, indicating that differences between higher and lower areas involve a true difference in feature selectivity, not only a difference in spatial selectivity.

The second variance partitioning analysis revealed that overall model accuracy reflected contributions from multiple feature subsets. Across all areas, the energy-mean features explained the largest proportion of the variance. In early areas, variance was also explained by pixel features and by linear-cross-orient features, while in more anterior areas, variance was explained by the linear auto features and the features computed by higher-order statistics of the steerable pyramid magnitude features. They also averaged the unique variance across voxels in each ROI, repeated the analysis and found again a primary role of the energy-mean feature, in all ROIs, as the unique variance explained by this feature was significant at the subject-averaged level in all ROIs except EBA, and in all subjects in V1, V2, V3 and hV4. The pixel feature made contributions in V1 and V2 especially, the linear cross oriented feature made a significant contribution in all ROIs at the subject averaged level, the linear auto features contributed in V3, OPA, PPA, RSC, FFA and EBA.

Through PCA, they identified a set of weights in voxel space that project the unique variance values onto a lower-dimensional space. Visualizing these weights for the first 4 principal components revealed several large-scale organizational motifs across the visual cortex. These motifs can be recovered based only on the patterns of unique variance across the feature subsets of the model, and texture statistics features may therefore provide a form of scaffolding for the emergence of higher-level semantic representational axes in the brain.

21 Natural Scenes reveal diverse representations of 2D and 3D body pose in the human brain

21.1 Reference

https://doi.org/10.1073/pnas.2317707121

21.2 Summary

While previous studies have identified cortical areas selective to body and body parts, the neural basis underlying the visual perceptions of body parts relationships has received less attention. By employing ground truth annotations and 3D-pose reconstruction algorithms, the authors of this study compared similarity patterns from cortical activity with similarity patterns from human pose models with different levels of depth availability and viewpoint dependency. They achieved significant correlations between pose models and cortical activity patterns and found that 3D view independent pose models, compared with two-dimensional models, better capture cortical areas activation.

21.3 Methodology

3D pose reconstruction: To reconstruct the 3D body pose from the images (they picked only the 4450 images containing bodies) the did the following:

1) they used a 3D human mesh reconstruction algorithm to extract 3D body meshes.

2) Meshes were parameterized by 3D joint rotation angles for major joints only (excluded jaw, fingers and toes), the algorithm also returns 3D joint rotation and 3D global rotation parameters.

3) Reconstructed 3D meshes were used to compute joint position and body orientation.

4) These information were used to measure dissimilarity between images.

Pose model and RDM: Annotations from the COCO image dataset (the dataset containing the natural images that were used as stimuli for the NSD) and 3D reconstructions were used to build the RDMs. They built 4 different ones: a view-dependent 3D pose RDM, a view-independent 3D pose RDM, a view-dependent 2D pose RDM and a viewpoint RDM.

Searchlight based Representational Similarity Analysis: It was performed with each of the 4 RDMs to identify different pose representations across cortical areas. It was also performed on 7 other control models. They first averaged and normalized Beta maps (provided by NSD data release). Neural RDM was derived using 1-correlation as the distance metric. for each searchlight neural RDM was correlated with normalized target RDM and correlation values were assigned to the central voxel of each searchlight, which resulted in correlation maps for each model for each participant.Correlation maps then were Fisher z-transformed and entered a group-level analysis to test the individual maps against 0 using a one-tailed t test at each voxel. Then t maps were corrected using a cluster-based non parametric Monte Carlo permutation test.

21.4 Results

They found a Distributed Cortical Network sensitive to the Geometric structure of poses. It includes areas in the lateral-occipital temporal cortex, in the fusiform gyrus, in the parahippocampal gyrus and the posterior parietal cortex, possibly including areas in the posterior superior temporal sulcus. Significant clusters were also found near V1, V2 and V3, from both 3D viewpoint and other pose models. Importantly, they found that activations from pose models extended into parts of EBA and FFA.

They conducted 2 partial correlation RSA analyses on the 3D pose models, the first on the view-dependent one (regressing out the view-dependent 2D pose RDM) and the other one the view-independent one (regressing out all the other models). As a result of the first one, they found that 7 out of 8 subjects showed sensitivity to depth information in the LOTC, while as a result of the second RSA they found clusters sensitive to intrinsic 3D pose information in the LOTC as well as in the pSTS. These 2 areas specifically might therefore encode intrinsic 3D pose info independent of the specific viewpoint.

They conducted RSA on other 7 control models, including HMAC C1, body polar angle, distance and size control models, as well as 3 models to control for mid and high level features. The results of this analysis suggest that pose models do capture cortical sensitivity to poses rather than merely low- or mid-level features.

22 Conclusions and Summary

22.1 Overview:

As demonstrated by the reviewed studies, the Natural Scenes Dataset (NSD) has been employed in a wide variety of projects involving all sorts of methods and topics. Researchers have used it for traditional encoding analyses, for state-of-the-art generative modeling, for semantic decoding, and for network-level investigations. A very wide range of methods and approaches have been applied to the NSD, ranging from more experimental and theoretical ones to more computational and engineering ones. This much variability can be observed when it comes to topic as well, as the studies presented analysed processing of visual inputs, recognition, temporal memory, differences in the visual pathways, semantic decoding, generative decoding, network topologies and more, Therefore, the most striking evidence is in itself the variability and divergence that can be observed between them, as it highlights the versatility of the NSD.

22.2 Methods and Approaches:

The diversity of methods applied to the NSD is one of the most prominent features of the literature. Studies employed a variety of methods such as voxel decomposition and matrix factorization to identify selective neural populations, classical encoding and decoding pipelines to link images with fMRI responses, and increasingly sophisticated and modern AI-based approaches such as diffusion models, graph neural networks, AlexNets and other pretrained models were combined with classical statistical analysis approaches for all sorts of purposes. A great variety of tools like CLIP, BigGAN, other generative frameworks, autoencoders and more were used to generate, reconstruct, decode or analyse visual and semantic information.

22.3 Topics:

Several recurring themes emerge from this body of work. More neuroscience-focused studies uncovered and analysed specific neural selectivities (e.g., discovered food-selective regions, created orientation maps, analysed category-selective regions and visual processing pipeline), examined representational dynamics over time (e.g., observed drift in V1 representations and how memory consolidation happens thanks to the MTL) and explored visual and semantic pathways as well as large-scale networks topologies (e.g., dorsal vs. ventral pathways contributions to objects and motion processing, discovery of the SAL/PMN integrated and largely distributed network and of amygdala's role in social cognition). At the same time, more engineering-focus studies employed the NSD to produce methodological advances in personalization of encoding models and synthetic stimuli (ensemble encoding models), in visual decoding and in semantic decoding.

22.4 Limitations and Conclusions:

Despite its richness, some limitations remain. First of all, training advanced models on NSD-scale data has high computational demands, and this of course is a factor restricting accessibility. Secondly, while the dataset is with no-doubt longitudinal and large, it still covers only eight participants, and this raises questions about population-level variability. A last, relatively important concern is linked to the fact that the NSD takes stimuli solely from the COCO dataset, and given the fact that the COCO dataset is primarily designed to be a computer vision resource, relying on it as a single image source might constrain ecological diversity, as it privileges certain categories over other, equally important ones. However, even when considering these few limitations, the NSD is unique thanks to its variety of tasks involving multiple processes, its scale and its resolution. Beyond this, the fact that the dataset is delivered with preprocessing, population receptive field estimates, and ROI definitions, allows for lowering technical barriers and enables access for research communities coming from different backgrounds. Because of these reasons, this dataset is most likely one of the most powerful, versatile and unique resources currently available for studying the visual system and for many other applications.